



Real-Time Basketball Scoring & Player Recognition

Product Requirements Document & Roadmap

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1. Product Overview

1.1 Problem Statement

Pickup and leisure basketball games have no easy way to track scores per player and per team in real time. Existing solutions either require expensive hardware, save full game footage to third-party servers, or only provide post-game analysis. Players are left keeping score manually or relying on memory.

1.2 Solution

CourtIQ is a mobile-first application that uses a single iPhone on a tripod to:

- Recognize each player's face in real time using on-device AI
- Automatically detect when a basket is scored
- Assign points to the scoring player and their team live
- Display a live scoreboard (individual + team) on screen
- Save only scored moments as short clips — never the full game

1.3 Vision Statement

Give every pickup basketball player the analytics experience of a pro — privately, affordably, and in real time.

1.4 Target Users

User Type	Description	Core Need
Pickup Player	Casual player at local court, park, or gym	Know who scored what, settle disputes
Rec League Organizer	Runs regular weekly games with consistent teams	Per-game and season stats per player
Friend Group	Same group plays weekly	Leaderboard, bragging rights, highlights

2. Goals & Non-Goals

2.1 Goals

- Real-time face recognition — identify registered players from a live camera feed
- Dual scoreboard — track individual player points AND team totals simultaneously
- Auto score detection — detect when a ball crosses the rim and increment scores
- Privacy-first — no full game video stored; only scored moment clips saved locally
- Works on a single iPhone 13 Pro or newer on a tripod — no extra hardware
- Player profiles — face embeddings + stats stored per player, not raw photos

2.2 Non-Goals (MVP)

- Auto pan/zoom camera tracking (Phase 2)
- 3-point vs 2-point automatic detection (Phase 2)
- Cloud sync and cross-device access (Phase 2)
- Full game video recording and upload (Phase 3)
- Public leaderboards or social sharing (Phase 3)
- Android support (post-MVP)

3. Feature Requirements

3.1 Player Registration

Before a game, each player registers their face in the app.

Feature	Description	Priority
Face capture	Capture 3–5 face photos per player via front or rear camera	P0
Embedding extraction	Extract numeric face embedding vector on-device (Apple Vision)	P0
Player profile	Store name, team colour, embedding — no raw photos retained	P0
Team assignment	Assign player to Team A or Team B during registration	P0
Edit/remove player	Update or delete a player profile before or after a game	P1

3.2 Live Face Recognition

During a game, the app continuously identifies players appearing in the camera frame.

Feature	Description	Priority
Real-time detection	Identify faces in live camera feed at minimum 15fps	P0
Confidence threshold	Only confirm identity above 85% confidence score	P0
Unknown player flag	Flag unrecognised faces without crashing or assigning points	P0
Multi-face support	Track up to 10 players simultaneously in frame	P1

3.3 Score Detection & Attribution

The core feature — detect a basket and assign points to the right player and team.

Feature	Description	Priority
Ball detection	Detect basketball in frame using YOLOv8 model	P0
Rim detection	Identify basket rim coordinates via one-time court calibration	P0
Score event trigger	Trigger score when ball trajectory passes through rim zone	P0
Player attribution	Assign score to last recognised player who had possession	P0
Manual override	Allow tap-to-correct if auto attribution is wrong	P0
Point value	Default +2 per basket; manual toggle for +3 (Phase 2 auto)	P1

3.4 Dual Scoreboard

Live scoreboard visible on screen at all times during a game.

Feature	Description	Priority
Team score display	Show Team A vs Team B total points prominently	P0
Individual score display	Show each player's personal points tally	P0
Score animation	Visual flash/animation when a basket is scored	P1
Game timer	Optional countdown or count-up timer	P1
Undo last score	Remove the last attributed basket in case of error	P0

3.5 Clip Saving (Event-Triggered)

Video is only written to storage when a score event is detected — never the full game.

Feature	Description	Priority
Circular buffer	Keep rolling 15s video buffer in memory at all times	P0
Clip trigger	On score event: save 10s before + 5s after to local storage	P0
Clip metadata	Tag each clip with player name, team, points, timestamp	P0
Clip viewer	In-app gallery of scored moment clips per game	P1

Feature	Description	Priority
Highlight reel	Auto-compile all clips from a game into one video per player	P2

4. Technical Architecture

4.1 Tech Stack

Layer	Technology	Rationale
Frontend / UI	React Native (Expo)	Frontend dev familiarity, cross-platform future
Camera access	react-native-vision-camera	Frame processor support for real-time CV
Face recognition	Apple Vision Framework (via native module)	On-device, private, fast
Ball detection	YOLOv8 nano — CoreML converted	Lightweight, runs on iPhone GPU
Ball tracking	ByteTrack + Kalman Filter	Handles occlusion and fast movement
Score logic	Custom JS engine (event-driven)	Simple state machine for possession + rim event
Video buffer	AVFoundation circular buffer (native)	Efficient memory, no disk writes until score
Local storage	SQLite via Expo SQLite	Player profiles, embeddings, game history
Clip storage	iOS local file system (no cloud)	Privacy-first — stays on device

4.2 Data Model

Entity	Fields
Player	id, name, teamId, faceEmbedding (vector), totalPoints, gamesPlayed, createdAt
Team	id, name, colour, playerIds[]
Game	id, date, teamAId, teamBId, winnerId, duration, clipIds[]
ScoreEvent	id, gameId, playerId, teamId, points, timestamp, clipPath
Clip	id, scoreEventId, filePath, duration, thumbnailPath

4.3 Score Attribution Flow

The logic for assigning a basket to a player follows this sequence:

- Camera frame processed every ~66ms (15fps minimum)
- Detected faces matched against stored embeddings above confidence threshold
- Ball position tracked frame by frame via YOLOv8 + ByteTrack
- Last player within 2m of ball (by position estimate) flagged as possessor
- When ball trajectory crosses rim zone → score event fired
- Points (+2 default) assigned to flagged possessor and their team
- Circular buffer flushes clip to disk; metadata written to SQLite

5. Product Roadmap

Phase 1 — MVP (Weeks 1–4)

Goal: Working real-time scoreboard with face recognition. No video saved.

Week	Focus	Deliverables
Week 1	Foundation	Expo project setup, camera feed via vision-camera, Supabase/SQLite schema, player registration UI
Week 2	Face Recognition	Apple Vision face embedding extraction, player registration flow, real-time face matching in live feed
Week 3	Score Detection	YOLOv8 ball detection on iPhone, court calibration (rim coordinates), score event trigger logic
Week 4	Scoreboard + Polish	Live dual scoreboard UI (team + individual), manual score override, undo, basic game session management

Phase 2 — Enhanced Camera (Weeks 5–7)

Goal: BallerCam-style experience using just an iPhone.

Week	Focus	Deliverables
Week 5	Auto Zoom	AVCaptureDevice zoom API — auto-zoom toward ball/action zone
Week 6	Ball Tracking	ByteTrack + Kalman Filter integration for smoother tracking across frames
Week 7	3-Point Detection	Detect shot origin distance from rim to auto-assign +2 or +3 points

Phase 3 — Highlights & Clips (Weeks 8–10)

Goal: Scored moment clips auto-saved. Highlight reel per player per game.

Week	Focus	Deliverables
Week 8	Circular Buffer	15s rolling video buffer in memory; write to disk only on score event

Week	Focus	Deliverables
Week 9	Clip Metadata	Tag clips with player, team, points, timestamp; in-app clip viewer
Week 10	Highlight Reel	Auto-compile all clips per game into a player highlight video; share sheet

Phase 4 — Growth (Post Week 10)

- Season stats and historical leaderboard across multiple games
- Android support (React Native makes this viable)
- Optional cloud sync for teams who want cross-device access
- Social sharing of highlight clips
- Rec league mode — scheduled games, standings, season rankings

6. Success Metrics

Phase	Metric	Target
MVP	Face recognition accuracy (known players)	> 85% in outdoor lighting
MVP	Score detection accuracy	> 80% of made baskets correctly attributed
MVP	Latency from basket to scoreboard update	< 2 seconds
Phase 2	Auto-zoom responsiveness	Ball in frame > 90% of play time
Phase 3	Clip save success rate	> 95% of score events produce a saved clip
Phase 3	False positive score events	< 5% of detected events are non-scores

7. Risks & Mitigations

Risk	Impact	Mitigation
Poor outdoor lighting degrades face recognition	High	Tune confidence threshold; add optional flash mode; prompt user to register in similar lighting
Ball moves too fast for detection at 15fps	High	Use TrackNet heatmap model as fallback; increase frame rate to 30fps if device allows
Wrong player attributed for a score	Medium	Manual override tap always available; undo last score button
iPhone overheats running CV + video simultaneously	Medium	Profile and throttle YOLOv8 to GPU; run face recognition at lower frequency (every 5 frames)
Rim detection fails on different court setups	Medium	One-time court calibration step before each game; allow manual rim zone adjustment